Simulation and Performance Analysis of TCP Congestion Control Algorithms using NS-3

Deadline: 24/6/24 & 21/6/26

Consider a real-world scenario of Intelligent Transportation System involving a Vehicular Ad Hoc Network (VANET), where multiple vehicles send traffic information, camera data, and road-condition updates to a traffic management server through Roadside Units (RSUs) or wireless infrastructure.

Analyse how different variants of TCP - TCP Reno, TCP Cubic, and TCP BBR work under network congestion and compare the performance WRT Throughput, Delay, Packet loss, Fairness and Congestion window for Reno, Cubic and BBR.

Justify why Low delay and reliable communication are important for intelligent transportation.

Vehicle 1 —|

Vehicle 2 —|

Vehicle 3 —-| — Roadside Router — Internet — Traffic Control Server

Vehicle 4 —|

Vehicle 5 —|